

AI FDE Challenge

AI FDE Observability Challenge — From Black Box to Explainable Production System

Participant Mission

You are the AI FDE team responsible for taking over a production AI decision system with weak observability.

Using the provided brownfield repository and AI-assisted prompts, your goal is to understand the system, expose its blind spots, instrument it, investigate a real incident, harden its observability, and demonstrate that it is production-ready.

Your tasks are:

1. Discover the End-to-End System & Business Flow Understand how a business request moves through APIs, services, data, AI/model components, decision logic, and downstream systems.
2. Assess the Current Observability Baseline Identify what logging, metrics, tracing, AI telemetry, business events, and audit evidence already exist—and what is currently invisible.
3. Identify Observability Gaps & Production Risks Find blind spots that could prevent the team from detecting failures, diagnosing incidents, explaining decisions, controlling cost, or proving compliance.
4. Define KPIs, SLIs, SLOs & Error Budgets Translate business and reliability objectives into measurable indicators, service targets, thresholds, and acceptable failure tolerance.
5. Design the Telemetry & Correlation Model Define how Trace IDs, Span IDs, Case/Transaction IDs, timestamps, service identities, model versions, and business context should connect across the system.
6. Implement End-to-End Distributed Tracing Instrument the complete transaction path so a single business case can be traced across API, application, AI/model, tools, data stores, and downstream dependencies.
7. Harden Structured Logging & Protect Sensitive Data Improve log structure, correlation, and diagnostic value while identifying and removing unnecessary PII, secrets, or sensitive business data.
8. Instrument AI Quality, Usage, Cost & Agent Behaviour Capture model/version information, latency, token/compute consumption, retries, fallbacks, errors, quality indicators, and relevant AI execution events.
9. Create a Unified Production Health View Design an operational view combining reliability, performance, AI quality, cost, business outcomes, security, risk, and compliance signals.
10. Investigate the Production Incident & Reconstruct the Timeline Use logs, metrics, traces, AI telemetry, and business events to determine exactly what happened to the affected transaction.
11. Determine Root Cause, Critical Path & Business Impact Identify the component or behavior responsible for the incident, determine the performance critical path, and quantify the impact on the customer and business.

12. Remediate the Observability & Production Weaknesses Implement the smallest production-safe changes needed to close the identified gaps without unnecessarily changing business functionality.
13. Design Alerts, Thresholds, Escalation & Runbooks Define when the system should alert, how severity should be determined, who owns the response, and what operational steps should be followed.
14. Establish Decision Provenance & Audit Evidence Ensure important AI decisions can be reconstructed from input and context through model/tool execution, controls, human intervention, and final outcome.
15. Execute the AI FDE Production Readiness Gate Determine whether the system can reliably detect failures, trace transactions, measure AI quality and cost, recover safely, and prove that controls operated.
16. Validate Recovery & Produce the Final Evidence Pack Re-run the problematic scenarios, demonstrate that the fixes work, verify SLO recovery, and produce the evidence needed to justify production readiness.

The overall challenge flow

Discover → Baseline → Identify Gaps → Define Objectives → Instrument → Trace → Observe → Investigate → Diagnose → Remediate → Alert → Prove → Validate

The key outcome is not simply “add observability tooling.” It is:

Transform a production AI system from a black box into a system that can be observed, traced, diagnosed, explained, recovered, and audited with evidence.